

PRESSURE MEASUREMENT REQUIREMENTS — v1 GAP ANALYSIS

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1. Purpose

The v1 board was designed from a sketch without written requirements. This document works out what a pressure channel on a solid rocket motor test stand actually needs, and compares that to what v1 can do.

2. What the test needs

During a static fire, the pressure inside the motor chamber is the primary measurement. Student solid motors typically run at 300–800 psi, with a sharp rise at ignition lasting a few milliseconds.

- **Range:** 0–1000 psi, to cover peak pressure with margin
- **Sensor type:** an industrial pressure transducer with a 0.5–4.5 V output, rated for the chamber's heat and vibration
- **Sample rate:** at least 1000 readings per second, to capture the ignition spike
- **Environment:** long cable from the test stand to the DAQ, near an ignition system that produces electrical noise
- **Connections:** secure, screw-down wiring that won't shake loose

3. What v1 can do

v1 uses an I²C sensor connection. Small I²C pressure sensors (BMP280 class) measure roughly 0–17 psi at about 100 readings per second, over a wire run of a few inches.

4. Gap

Requirement	Needed	v1	Meets?
Pressure range	0–1000 psi	~0–17 psi	No
Sample rate	1000/s	~100/s	No
Cable length	several meters	inches	No
Noise protection	filtering + input protection	none	No
Wiring	screw terminals	loose socket	No

5. Conclusion

v1 works as an Arduino sensor carrier but cannot measure chamber pressure. It is suitable as a bench prototype for testing the data path. A version 2 built around a 0–1000 psi transducer with

filtering, input protection, and screw terminals is needed. The v1 board will be built and tested first to validate the data logging pipeline.